ActInf Textbook Group ~ Cohort 7 ~ Session 19 (Chapter 7) ~ 3/13/2025

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all right welcome back
everyone we're in cohort 7 chapter
so for some of us this will be our first
time on seven others not not so
first um but where would anyone like to
begin with a specific quote or question
about seven or how did anybody take
seven or where do they see seven in the
book
maybe I'll say uh I I listened to the uh
video of the last session the evening
session where Andrew talked about
chapter 7 and so what interests me uh
and I'm still struggling how to code
things like free energy principle and
how to um you know how all the math
Works Etc but um and he did he did help
explaining that but what I got from
these sections was that they
progressively um are looking at modeling
different parts of mental life um and
showing that in each case you know the
this apparatus um uh can be used to
model that and so in my case uh like so
basically like perceptual processing
like deducing an intended signal uh the
teamas with the rat like decision-
making planning uh epistemic versus
pragmatic and then information seeking
with the isaad uh seeking position where
you can get it kind of yielding a street
light effect and then learning a novelty
so this synthetic worm exploring its
environment changing the generative
model and then finally like hierarchical

or deep inference words and versus
sentences putting that together together
on separate time scales so so what it
seems like is that
um you end up you you know you you see
oh I could use active inference for
modeling this um and then um you do
things of you know whatever
sophistication you can and so I'm
interested maybe in the opposite to say
okay I'd like to have develop a
cognitive model I think that I'm going
to have like a continuous part and a
discrete part like a procedure ual
continuous mind and a discreet
declarative mind and then I'm going to
maybe Wonder like how they discover
critical points navigate them um you
know have a language of critical points
so so I guess I'm just still learning
about U the um active inference but I'm
trying to pick up on that connection
between like having a mental process or
experience in life you're trying to
model and then thinking okay where does
predict addiction error come in I guess
that's probably what it's about that's
where I'm at right now thank you awesome
summary anyone else want to give some
thoughts on seven or like a a section or
a quote or a figure or equation that
they want to jump
to um I mean figure 7.2 with the the
plots about uh The Listener are a little
confusing I guess um I'm not sure what
I'm seeing in the First Column of panels
there maybe someone with some more
insight could kind of walk us through

what these panels are
showing yeah it's a great point because
this chapter is the discret setting but
then it's like wait a minute why are
there continuous
traces here with a discretized model so
these are kind of like some model
outputs from the
SPM mat lab
scripts um but yeah we we can definitely
get and also it doesn't help that
they're black and white yeah so it's a
little bit like well but are we expected
to I mean how can you
trace so I agree they're they're not
super
informative there do we know at least
like what they're meant to be showing
like what is yeah I think like I could
say the main idea if you show the
diagram there if you go up to the
diagram the main idea is that there's an
amateur musician kind of playing some of
the notes wrong and um but if you go if
you go to the if you go to the screen
someone's listening and they kind of
know what the melody kind of should be
it should just go dun dun dun dun and
then D so when the musician misplays it
which would be the lower right
they they misplay it but the person who
has the idea of what the melody should
probably be like they interpret it well
it's gonna it really was meant to be
that it's the intended signal as opposed
to the actual signal and so active
inference is able to you know if you
have a strong enough Matrix where the

pattern is so insistent that you know
it's going to survive the truth uh then
um then it can model that so yeah okay
and what's happening with these figure
specifically so top left posterior
probabilities about each note so what's
happening here is basically how
confident are we each of these is a
different note but they they can't be
determined so it's like in time point
one and also does it doesn't help that
it's kind of offset yeah here but it's
sort of getting at they being like the
most likely one is on the top
so that's one with the highest posterior
probability which is kind of selected if
you only get one
selection and then on the lower left is
the the prediction errors over time so
like you can see there is a higher
prediction
error when the incorrect note is
played okay and but each line though is
is what it's a it's a posterior
probability about a note yeah yes so
it's like the the the Laten um there's
one spot for what's the true note being
played like we're going to receive a one
hot observation like we either hear um
note one two 3 or four that's coming in
with a vector that looks like 1 0 0 0
for the first note y like in this
situation then we have a prior belief
distribution over those four so we could
have a uniform belief over the four 2525
you know or you could have um
333 sure okay so then the relationship
between the skewedness of your prior and

the attention to the sensor input is
what dictates belief updating which is
when the prior becomes the posterior so
in the extreme case of no attention it's
like the observation didn't even come in
you just carry your prior forward in the
extreme case of total attention you just
basically update your posterior to
whatever the observation was and then
everything in between is
like attention weighted belief
updating and so there there could be
some value of attention where this is
exactly what would be
heard but in that point you don't even
need to do basian statistics you just do
descriptive statistics on incoming
information right yeah or or the
converse so at what what when you're
looking at like the trace of your
simulation at each time point you have
the belief distribution over possible
notes so again even though it's only
like one latent State
Dimension it's a distribution it's a
categorical distribution over four
options and so this is kind of like
plotting the again we we could replic
and color these lines to trace them more
clearly but like we can see that in this
S3 moment here's where it's sort of most
mixish like there's just a little bit of
weird stuff happening but to investigate
it more we we'd want to
um run it but this is like this and
that's why it's built up this way
because again chapter six is kind of the
recipe okay so just just to be clear so

on the on the plot
so like let's take one of these lines
for example say that the Top Line um
yeah that one right there that is a
curve so this is going to correspond to
the probability the posterior
probability that it's I don't know note
one and then there's a corresponding
line at the bottom where that where
those where they
bifurcate right is that the so like uh
at the split around like. three do you
see that on the I am really not sure
because there's there's a flatline at
zero and one which might just be like
graphical yeah I think it's yeah then
there's one two three four five six
seven eight notes yeah well there's but
there's four notes right right so what
I'm not sure what the eight lines are so
I is it like is it like one of the the
lines corresponds to the posterior
probability that it's not the note that
we're talking about
you know I I'm not sure what it's what
it's meant to be showing I agree and I'm
not sure if if there's eight notes like
in the scale and they're like tending to
zero here which is why at the end but
then why why would you have high
posterior probability belief in multiple
notes at the end right I mean I mean it
it looks like they're meant to be suming
to one so I think it it would have to be
like like one like two of the like the
lines come in pairs basically I think
because otherwise it's not summing to
one right cars like a upper and a

lower and maybe they but they don't sum
to one uh they don't sum to one if you
look at the third column they definitely
don't sum to one but then it's not a
probability distribution what what what
they look like uh they look like error
bounds like from below and from above
like how confident are you that you hear
what you that you hear what you're
hearing right and and somehow like with
the third one the upper confidence just
drops like uh I don't think you know
we're not sure we can't be sure because
there's a goof here so yeah
it again we should investigate it more I
agree that this is not super clear also
it could be if it's a posterior
probabilities about each note then it
could be tra there could be eight notes
possible yeah but then it's like um you
know yeah yeah like at this moment it's
like there's an 80% chance it's Note One
70% right but then so that doesn't sum
to
one because that's I was thinking yeah
that's why I was thinking like if you
take that that 80% example there might
be a corresponding line down below
that's around like the 20% that's
supposed to be like the like the not
that note line you know so it would some
to one but is that not what it is yeah I
I I agree and also so just again the
fact that it's continuous yeah and it
has like different Dynamics it's just
it's part of the trace of mat
lob okay in the discreet
setting this doesn't really come into

play but it's yeah I I agree
like it's not very
clear okay so I'm not alone in that
that's good to know yeah and and
especially just because this kind of
gradient method The
Continuous um gradient subtime step
scale gradients which can happen but
that's not the calculations that are
needed for discrete calculation which is
just a basian belief updating which is
just done in one shot in this
case um but yeah
like this
chapter the first two examples are kind
of
like two of the core Tools in thinking
about perception and action
models it's they're really really
critical just to structurally
understand that there's observations
that are emitted from a latent State
there could be seven dimensions of s
projecting onto one o or it could be the
other way around any configuration and
there's some intermediator a that is
able to take a latent State
distribution and project it onto an
observation distribution and vice versa
so that's the sort of bacing nature of a
that it can be used in its generative
capacity or in it recognition
capacity you need a Laten State prior to
kick this whole thing
off um and it's usually shown with
three but really it's just
two it's really just you kick it off and
then you're only looking at one

transition
step and then where you transition to
becomes the the prior Distribution on
the next moment so even though there is
a a trace of observations
we don't model observations as causing
successive sequential
observations we model changes happening
in an unfolding Laten space that emit
observations and then action enters the
picture in choosing the transition
operations in the latent State like
those are sort of key pieces that
are at a very high structural level
important for this type of modeling
Anders and those latent spaces are kind
of States I mean are generally kind of
imaginary like it's right I mean like
that's partly it's it's usually not
necessarily like exactly real but it's
more like that's what's that's what's
either modeled um by the modeler or
modeled by the
organism but yes is that Bally a good
way to look at it like they're not
they're not intended necessarily to be
real let's say it's a temperature
situation the agent's model is like the
the true underlying temperature but it's
the agent model of the true temperature
and the observations they're getting so
it's not the generative process how it's
described in the book of like the quote
real temperature in the
simulation mhm so yes this is like these
are all variables with in the agent's
cognitive
model and then there's another which is

also our model because we're modeling
the agent it's like a okay we're both
modeling together as friends yes and
that's that is brought up in chapter
nine that's the metab basian
perspective which is we are outside of
the agents generative model proposing
that as a map of a cognitive
territory which is itself mapping a
reference outside of itself um but
there's another degree of mod freedom in
like saying well like um there could be
aspects of the environment of the
agent that the agent doesn't even know
about so that that and I
think I think that that's a key thing
for discret language like how why
discret langu anguage works at all it's
because different people modeling you
know basically a similar situation are
going to come up with similar answers
even though they're you know even though
those latent states are all not real but
but they're real in the sense that
you're going to get similar answers and
so um and so and they're going to get
discreet answers so you found the
similar critical points you know you
found so I think that's that's the
Wonder of a discret communication
yes that tension with continuous
discreet comes up in in so many guises
both in the treatment of time and in the
treatment of the state spaces
themselves so here we have in chapter 7
basically a focus though with some some
spice added discrete State space
discreet time so again confused by these

Trac
is there's not necessarily
differentiability there's just snapshot
based discre updating in the continuous
State space um how but the advantage of
the the
discretization is first off it's very
easy to understand the computational
resourcing and when it comes to action
selection it's easy to roll out
counterfactuals like chess Trac
search chapter 8 brings in the
continuous time and continuous State
space
mathematics but yeah where we can go to
another section or we can look at um the
specification in
PDP um or I wanted to ask is there an
open source like a version of mat lab
like o octave will that open work these
files or not there's a standalone
download of SPM that runs without a mat
lab license there is octave which is
like a patched open source mat
lab I've never run SPM in
octave so that's why in the sort of post
2018 is period there was movement just
away from at laab but it still is used
by many people who and many models that
are in there but um you know of course
any language can can emulate any
language but using Python and
Julia and a lot of other uh language
it's taking us closer more to the the
data science pipelines that are and the
other packages that are
relevant so I would say the best place
to look for this is in the pmdp

documentation like that's why it's
called Pi
mdp pmdp is not trying to be a generic
model of all base graphs kind of in this
in the spirit of RX and
fur the agents is literally defined
almost exactly according to the
textbook the same tas's
situation but this is exactly
how it's laid out with agents and the
ABCDE okay nice it just goes through
them just there you know categorical
Distribution on
a b
Matrix that's how you encode the
movement
preference prior on hidden States d
and here ABC
D
are
specified and then when you all there is
to run and and pdps have been around for
decades they're not always fit using a
variational or an expected free
energy but this is basically what the
loop
is you get
the observation from in in this case
from the location in the grid
World belief updating on the latent
State what is my true location a curse
so that's inference of the latent States
given the observation index and the a
matrix and your prior Distribution on
latent
States so that's belief
updating that's the sense making that's
the perception that's the inbound that's

the music listening then there's going
to be action
selection first expected free energy is
calculated for each possible action so
that's for the given time Horizon it's
going to ask if I were to do that
sequence of events whether it's just one
action like one time step planning or
whether it's multiple time steps of
planning in the discreet case
each of those chains of actions will be
unrolled in terms of their expected
effects on latent States via the B
Matrix and then those get unfolded into
the expected observation
distributions those expected observation
distributions are evaluated in terms of
their epistemic and pragmatic value
pragmatic value how well would those
observations align with my preferences
epistemic value what would be the
expected Information Gain from those
observations those scores constitute the
G Vector which lines up with your policy
options those scores in G which don't
have like an intrinsic
meaning are used to reweight the policy
prior into the policy posterior so
policy prior is what you're believing
that you're apt to do before the
observation gets re ated according to
expected free energy values relatively
and renormalized with a softmax so that
it stays as a probability distribution
over
actions one action is chosen from the
policy posterior
distribution that is going to be the

selected action and
then at the end of this cycle the
um it's it kind of click ahead to the
next Laten State belief and then that
will be used here in the next cycle as
beliefs about
States so that's one procedural way to
have the sort of imperative style for
describing a perception action
Loop but there's other orderings that
could be possible
like you could you could do different
orderings I'll just say that so that's
why there's a lot of degrees of model or
freedom because this is not it's not
like this is the only way to
model the handling and that's even
within this simple single observation
single action
setting but that's how it works in pmdp
which is basically what is described in
the
textbook nice that's super helpful um
what did you mean by other orderings
orderings of of what
um like there's
um this is kind of a classic like
observation comes in update beliefs
choose an action based upon the updated
beliefs but for example you could have a
a a Time step situation where it's like
at the beginning of each time step the
agent chooses an action based upon what
it
believes and then gets an observation
and uses that in the next time step to I
see so that would be you know and then
especially if you're dealing with in

this example there's basically only two
operations
happening the state inference and the
policy inference so those are kind of
the two main categories of
orderings right which one do you update
first and it totally makes sense to
update the observation first
um as you get more kinds of variables in
play and more cognitive mechanisms there
become more ways to imperatively specify
The Ordering of the loop right like if
there was a read and write to memory if
there was an attention
element if there was a policy search all
these different
features and then what's what's really
interesting and what kind of returns the
focus to to the high road and the low
road is
like the free energy principle is
describing the physics of flows across
these graphs not in discrete
time but it can be
discretized but as this sort of
unfolding simultaneous
flow in terms of belief updating on this
graph so and and the way RX INF fur kind
of aims to handle that is with with
reactive message passing sort of on
demand just in time node
updating but even then that's not
happening continuously it could be
happening rapidly or infrequently but it
it is happening in especially on digital
computers at some point everything is
going to be discretized in a way um
so figuring

out what the duty cycle is for the agent
and that's where you get like the
tote like this is sort of an older like
test operate test
exit this is one like you
know whatever it sort of seems like
esoteric or strange to be talking about
the ordering of perception action it's
like there's
tote and then of course there's the UDA
Loop which is another discretized way
and and that has been connected with
active inference as well so just when
you get down to building on the low road
which is what chapter seven and eight
and everything the whole second half of
the book is about you get down to like
these details about which variable are
exactly undergoing which operations and
which
orderings but that's not something that
the framework is opinionated on but it
is something that's required to have
operating
code yeah if so um I remember that like
one of the central not well to me anyway
at the time that I read it it seemed
like a central
tenant uh was that what active inference
sort of does was is is that it um sort
of like flips the way that an agent uh
perceives the world in that instead of
waiting for the observation to come in
and then making a prediction about what
they're seeing they make a prediction
first about what they're they think
they're going to see right and then they
take the observation and it's the

prediction error that like kind of
Cascades through the
system right so is is is that like a
different order that that you're talking
about or is that uh kind of set in the
way that it
works does that question make sense yeah
yeah totally so like
um there's a few ways to go about this
if you have a sequence of alternating
you know belief updating and then action
updating you know Laten belief updating
action State
updating at some point which one
happened first right just alternating
and so it it it doesn't really matter
but then when you think about the
injection of an observation that's kind
of what we were just exploring like
observation belief action observation
belief action was how the PDP had it as
opposed to Observation action you know
so once you get to to three or more you
do get different sequential possible
orderings as well as at the level of the
simulation where okay now we have the
like for example if we look at the open
AI gym
environment so this is a discret
environment where every time
step um whatever it's called now but
every time step like there's getting
gets an action sends back so that's a
kind of call and response style but when
it's one agent in one one environment
you could do that or you could have it
send two and then one back or something
like that

but um then you kind of point to like
this idea that that the agent is is like
action as fulfilled prediction
planning proceeds by inferring an
action what the
agent feels in Char nothing
but like can't understand
you your audio is cutting out a little
bit the the uh the
organism is in charge by making
predictions all the time like even if it
knows nothing then it predicts that it
knows nothing but typically it knows a
lot about its environment and it's uh
you know that that that knowledge is
stable and just it's reporting minor
changes to that you know within that
environment yes
like yes there's the sort of like inner
agency of like it's not just taking
punches from the environment even just
when it comes to sense Mak so just
thinking about the first 7.1
example each observation is meeting what
can be thought of as like an active
prediction but it's the prior the prior
is being brought to the table and
meeting the observation and that is the
basan updating moment so there's that
element which is it's not just
processing inbound information but
there's this sort of active listening
approach and then that goes further when
we actually
consider over
action especially action in terms of
what is expected to be
perceived and then the agent is

unsurprised when the consequences of
action are what it expects to be
surprise enters when the consequences of
action are not what was
expected so that kind of centers
action
in setting what is the envelope of
observations that would or wouldn't be
surprising but it because it's so
General we describe rocks and you know
other kinds of you know abstract um
boundaries it's hard to personify it or
apply too much of a psychological lens
to the generic case but in cases that
are able to have that kind of
psychological projection that's
exactly that's how we compose
expressions in the active inference
ontology to make structured models for
those
phenomena and and one difference uh that
comes up like in that so like in
cognition I think that's very much the
case where like you're always predicting
uh your so you can always be surprised
and you can always deal with that
surprise but in the case of emotion uh
you have
expectations and so if those aren't met
you may feel surprised or or if they're
met excited or sad or happy but in the
case of I would say like fright or
disgust you refuse to make expectations
you know there's certain things that
happen that you refuse to expect and so
then
somehow you're dealing with a different
situation

it's a different
game that's interesting what do you mean
um you you refuse to make an expectation
in cases where what where like
well so this is a this is a simple this
is a simple model of emotion but if I'm
have a theory you know and it's
confirmed uh I'm excited you know if
it's about the real world if it's not
confirmed I'm surprised oh what happened
um but and if things matter to me then
it's not just uh excited surprised or
I'll be happy or I'll be sad you know I
may be devastated but if something
happened on the outside and I didn't
expect it because it was too fast or too
weird or just
too some kind of I just didn't want them
have that expectation those are the
things that are frightening you see so
those are or vice versa if it's on the
inside and I just refuse to make certain
expectations or they're just too sudden
or they're too weird uh those are the
things that are disgusting so that's an
emotional model but but it's it says
that in the emotional World it works a
bit different than in the cognitive
world yeah the some of the citations and
work on emotions and and exactly that
like when the rate or type of
change of a primary
model um that is where certain emotional
phenomena could be
modeled and and but again how you
actually what phenomena you choose like
for if we were doing a body temperature
example it's like maybe if the first

derivative of surprise through time or
like the time rolling average of
surprise is low even if our body
temperature is going up and down our
anxiety level is low but then if there's
rapid
changes there's a second level model or
a model that's taking in as observation
the first or the second derivative of
the changes in Surprise and body
temperature that that could be
associated with an anxious
response and maybe just say like active
inference is active by Nature uh so in
these situations you may refuse to have
a generative model you know of certain
kinds because that's not the generative
model you want to live by you know or so
then when things go outside that you
just resolve to passive inference you
just throw up you just throw up or you
just uh you just lay down and die or
just you know you go into some kind of
reflex modes where there is no
generative model you're just acting you
know passively passive
inference I don't know if there's no
generative model in that situation yeah
a person looking at that a person
looking at that in in an observation
setting it's not like just cuz a you
know squirrel has a flight or fight it
doesn't cease to have a generative model
right um that well that's I no I guess
so the the thesis would be that in
certain situations you don't need a
generative model you just need to act
and you just act as your program you

don't need to learn you're just going to
you know you're just going to but I
think that is
model right it's just that you have
maybe different like a like a like a
refined or um confined set of actions
that you could take as a result of
whatever you're perceiving right yeah no
like for example like death like I don't
have a model of my death I don't think
about it it's not healthy to think about
death you know so typically like you
would not want to model certain things
like that at all you just don't touch
them then just model though you do have
a I don't know like a state that is a
reaction to death or thinking about
death or or in fact in your case the
state would be not thinking about death
right exactly so if I'm that's still
part of your generative model I
think well what I'm trying to say is
that there's something more to life than
just to generative model right there's
the territory I mean this is just
there's the territory this is the
accounting
ethology behavioral scientific approach
to modeling cognitive systems but it's
like yes there's an infinite number of
points that are sort of correlat like
that maps are not territories restated
in any number of important
ways like the structure of the
observations coming in from the
thermometer does not have to be anything
having to do with what the cognitive
model of the agent as set up by a given

individual is
MHM and and that's sort of The Refuge of
the instrument
anals because it's
like here's our proposal for how to
reduce our uncertainty about
this do you have another
tool what what other tools are in
trade-offs with this
one whereas the realist is actually
looking to to wrestle on that more
ontological reality
mapped dimension
about things as they
are rather than more of the
statistical
[Music]
um approach to modeling
perception and behavior as
perceived which is why but that's part
of this fascinating view from the inside
view from the outside
dialectic mhm because people will bring
up first person phenomenological
experiences and there's so much work on
neurophenomenology and cognitive
modeling From firstperson
perspective whereas
in Behavioral Science and entomology and
things like
that it isn't it's not waiting for some
kind of first-person
report and and and I think there's a
dialogue between the continual
continuous mind and the discreet mind
like there's a cedal continuous mind
that just does its thing
reflexively but it you know it looks

they're they're helping each other and
so there's this discret mind that's
managing this generative model and you
know basically is able to and somehow
they're able to do but sometimes the
discret mind just lets go and it just
goes into continuous mode wherever
things
tumble well that's sort of what is laid
out here but this whole thing is a
generative model so it's it's not like
there's any utility
as far as I'm seeing it just to disre
the whole thing is the generative model
and now whether it's done in a more
habitual
fashion or it engages in a discretized
tree based policy roll out or some other
proposal for like recurrent switching
between different
modes however it's accounted for that
accounting is the generative model
but it can engage in completely you
could say well I have a generative model
where at the top level there's a switch
and it's you know it's Counting Apples
and then it switches to counting
oranges I guess I guess the presumption
that I'm making is I'm drawing the line
not between an organism and its
environment but between the organs
organism's internal mind and then its
whatever mind it has that represents
that environment like the unconscious so
in a human being like it be the
conscious and the conscious there's a
gap that's what's being modeled with all
of this

framework um
so the unconscious is really just part
of the world so to speak and you're
trying to talk to your unconscious you
know manage your you know relate to your
unconscious perceive it and and and act
on
it yeah like the B matrices in the te's
example in chapter 7 it's like these are
the mouse's
location mhm can intention upon it doing
what and then that in this case gets
just sort of faithfully enacted by the
environments but that part is not
emphasized here but then there's there's
situations like the windy grid
World which is a
classic um grid World kind of
variance where you have
um that kind of up down left right
and then there's wind that blows it like
zero or one or two
cells so in that situation or icy grid
world where it slides
occasionally so but but the agent's
model might not it might be like whoa
now I'm over here so it doesn't even
know about wind or ice it's just like oh
but I want to be expect to be here I
expect to get information if I go
here but yes
so that's something in the Triangle I
talked about the triangle uh language
geometry language I'm doing and then I
realized a good way to do it um that's
in this spirit is that the the
participant the agent or the triangle
Creator the S the one who's looking for

these triangle centers they only
understand triangles in terms of integer
lengths you know so it's basically like
rational numbers so they don't have a
concept of the real number so they're
trying to find a center which may not
exist as a rational number and so that's
the kind of like it's already in a
paradoxical situation where it's trying
to do something basically impossible so
it'll it'll do it as good as it can and
then it'll move on to another task let's
say so that's a nice so it's nice that
it can enter into the model that way
naturally oh yeah like let's just say
there was a super sharp preference for
to land on the number
10 um and but you had the policy you
could only move up or down by 50 to 80
so you were just sort of like
overshooting and then undershooting
overshooting and undershooting
um and then you could have another level
of the generative model that was about
task switching and there's different
ways to do task switching or like going
back to the the um the playing dead
example like you can either have um
stay as one of the actions in a one
layer model up down left right
stay or you could have a two-level model
where the top one is a binary decision
to move or stay and then within move
there's four
actions like those are all different
maps and again which map you make of the
city depends on what you want to do with
it who's funding you all these different

kinds of
questions what cognitive model you make
is contingent upon what you want to
account for what you what you um
do and don't want to be surprised
by and especially when it comes to these
cognitive functions and modeling
them
like where these models sit from mainly
taking in data even though the in in
theory these are totally just two two
directions in the same window especially
when we get to chapter 9 with the
empirical data analysis but
sometimes the model is set up to be
taking in data and the modeler is
interested in the parametric estimates
at the
end so to use use this framework to
model it's a human experience um
whatever I would find interesting I
think it seems like the crucial thing is
to discover well what is the prediction
you know and what is the error like if
because is that how I have to think in
terms of in order to this active
inference framework or are there other
Concepts I have to worry
about you don't have to worry about
anything I want to worry about something
um I I think it's relevant to look at if
we're thinking in discreet time but
figure 4.3 shows the the the
isomorphic nature of the continuous and
the discreet time generative model kind
of Meta Family
units again these can be nested composed
concatenated all those kinds of category

theoretic operations are possible but
this is the motif which is like what is
the unfolding State
space what observables are sloughed off
of that unfolding State space and then
what are the interventions in that
unfolding State
space so there's there's sort of
simplification cases where it's like
there's only one action so just it
unfolds according to just a Markov chain
without policy it's just a Markov
process and then you go okay well what
if there were um multi-way options like
in in the sort of wool from Computing
ontology in the multi-way compute where
there's like different bifurcating
possibilities from a given position then
it's a Markov decision process and then
if the observable is the chain unfolding
itself you don't need a partial
observability elements but then if you
want to model sensory ambiguity or some
kind of degenerate or or nonlinear or um
ambiguous
mapping between the true unfolding
Dynamics and observables to a given
measurement maker then you need
something like an a
matrix so if I could just interject
there for a moment there's not much time
left and I have to go at 9 speaking of
the a matrix can we go to figure 7.4 or
yeah yeah these matrices and just um
they're not too complicated but maybe we
can just like kind of go through each
one and just be clear what they are and
so um the caption anyway says I think

there's a type on the caption
right it says
um it says so each element of these
matrices
is the probability of the outcome
Illustrated at the end of the row
conditioned on the context being one and
being the location indicated by the
row so there the context being one here
is it would have been clear if it was
written as a one but there's two
contexts this is context one with the
food on the right this is context two
with the food on the
left so here flipping back and forth yep
we see
basically we see these two are
flipping and this 298 is flipping so
basically there's two a matrices in this
example there's what
context what context is one
in that's
A2 and then there's what location am I
in that's A1
okay see that's confusing I thought I
thought the context meant uh what the
rat saw the Q to
be yes a to be it's not the context is
where this thing actually is there's
kind of two there's two scenarios two
actual scenarios right when the context
is
unknown that's this
one okay so okay so let's let's read off
one of these columns just see if we can
understand it so like in in cell one and
A2 this is the probability that the rat
is going to be

in so I think it's a typo so it should
be that the rat is going to be in the
state indicated by the column
right given the the that there's no
context or that that what if I'm in the
start location okay I am sure that I
don't know the context don't know the
context okay
if you're in the bottom location I'm GNA
I'm GNA find out what the context
so then then you do know the context
then yeah that gets injected in when it
does go there when it discovers art so
then when it goes when it goes to the
bottom that this is where A1 comes into
play okay when I get to the bottom in
context one this is the difference
between these two y then I find out that
the Q is
R okay so it's the probability that you
find out that the that the Q is R when
you're when you so hold on sorry this is
observation and each observation we're
getting two features from the
environment this is the no food MH sorry
if I if I said this was the no context
earlier this is neither food nor not
food okay this is yes for food this is
no for
food so starting position I'm sure I'm
in the starting position and there's no
food bottom I'm sure I I'm sure that the
L that is telling me the Q is r and
there's no
food okay so let's go to row two that's
where something interesting is happening
this one
yeah yeah um if I'm on the left my my

belief is that if I go
left that there will be a 2% chance of
finding
food and a 98% chance of no
food if I go right vice
versa right but it's all given that the
Q is on the right that that the que
points to the right yes so so does the
rat know that the Q points right yes so
in this casee it would have had to it
have learned that that R means this and
L means
this okay so we assume so I think that's
what I was missing so it's already been
trained it's already been trained on the
que okay see I didn't okay I thought I
had to like go down first get the que
and then make a decision there's no
learning in this
model okay so it just so we assume that
the rat has seen or knows or learned the
queue exactly and this is the whole you
know Journey are we studying the
learning acquisition of the queue or are
we just looking at forward inference on
a behavioral trial given learning given
the learning okay thank you cool okay um
thanks all again like there was Daylight
Savings in some places but the place of
reference look at the UTC time in theod
to look at the calendar events the UTC
time is not changing the calendar events
don't change so that's where you'll see
everything so good thank you
Daniel bye